Program	Bachelor of Technology (BTech)	**Semester - 6**
Type of Course	-	
Prerequisite		
Rationale	-	
Effective From A.Y.	2020-21	

Teaching Scheme (Contact Hours)				Examination Scheme				
				Theory Marks		Practical Marks		
Lecture	Tutorial	Practical	Credit	T	T	P	P	Total Marks
4	-	4	6	100	60	-	50	210

***SEE** - Semester End Examination, **T** - Internal Theory, **P** - Internal Practical*

Course Content

T - Teaching Hours | **W** - Weightage

Sr.	Topics	T	W
1	**Introduction** Computer Vision Image Processing Low-level Computer Vision Mid-level Computer Vision High-level Computer Vision Overview of diverse Computer Vision	6	
2	**Computer Vision Applications** Document Image Analysis, Object Recognition, Tracking, Medical Image Analysis, Content-based Image Retrieval, Video Data Processing, Multimedia, Virtual Reality and Augmented Reality, Image Formation, Image Representation, Image Pre-processing Techniques	10	
3	**Feature Extraction** Point, line and edge detection, color, texture, shape and structure features in spatial and frequency domain, corner detection, Hough Transform	12	
4	**Image Segmentation** Gray-level thresholding, Supervised versus Unsupervised thresholding, Banalization using Otsu's method, Locally Adaptive Thresholding, Color-based Segmentation, Region-oriented Segmentation, Use of motion in Segmentation, Spatial Techniques, Frequency domain Techniques, Representation and Description - Boundary and Region Descriptors	12	
5	**Clustering** K-means, K-medoids, Mixture of Gaussians, Applications of Clustering	5	
6	**Classification** Discriminant Function, Supervised, Unsupervised and Semi-supervised; Classifiers- Bayes, KNN, ANN models, Applications of Classification	6	
	Total	51	

Suggested Distribution Of Theory Marks Using Bloom's Taxonomy

Level	Understanding	Application	Analyze	Create
Weightage	20	40	20	20

NOTE : This specification table shall be treated as a general guideline for the students and the teachers. The actual distribution of marks in the question paper may vary slightly from above table.

Course Outcomes

At the end of this course, students will be able to:	
CO1	**Understand the fundamental concepts of computer vision**
CO2	**Apply image processing and analysis techniques**
CO3	**Analyze low level computer vision using algorithms.**
CO4	**Build and train machine learning models for computer vision tasks**
CO5	Develop applications of computer vision in various domains

CO PO Mapping

CO	CO - 1	CO - 2	CO - 3	CO - 4	CO - 5
PO - 1	2				
PO - 2	0		3		
PO - 3				3	
PO - 4					
PO - 5					
PO - 6					
PO - 7			0		0
PO - 8					
PO - 9					
PO - 10					
PO - 11					
PO - 12					

Reference Books

1.	**Computer and Machine Vision (TextBook)** By Davies | Elsevier, Pub. Year 2012
2.	**Digital Image Processing (TextBook)** By R.C. Gonzalez, R.E. Woods | Pearson, Pub. Year 2018
3.	**Computer Vision- A Modern Approach (TextBook)** By A.F. David, J. Ponce | Pearson Education India, Pub. Year 2015
4.	**Think AI: Explore the flavours of Machine Learning, Neural Networks, Computer Vision and NLP with powerful Python libraries** By Swapnil Joshi Naik | BPB, Pub. Year 2022
5.	**Deep Learning with Python** By Francois Chollet | Manning, Pub. Year 2017

List of Practical

1.	**Display digital images using OpenCV**

	Write a program to read and display digital images using OpenCV. Become familiar with basic commands • Read and display image • Resize given image • Convert given color image into gray-scale image • Convert given color/gray-scale image into black & white image • Separate color image in three R,G & B planes • Create color image using R,G&B three separate planes
2.	**Point Processing Method** Write and execute image processing programs using point-processing method • Obtain negative image • Obtain flip image • Thresholding • Contrast stretching
3.	**Image Arithmetic Operations** Write and execute programs for image arithmetic operations • Addition of two images • Subtract one image from other image • Calculate mean value of image • Different brightness by changing mean value
4.	**Image Logical Operations** Write and execute programs for image logical operations • AND operation between two images • OR operation between two images • Calculate intersection of two images • Water marking using EX-OR operation • NOT operation (Negative image)
5.	**Histogram calculation and equalization** Write a program for histogram calculation and equalization
6.	**Geometric Transformation of Images**

	Write and execute programs for geometric transformation of image • Translation • Scaling • Rotation • Shrinking • Zooming
7.	**Image Restoration** To understand various image noise models and to write programs for image restoration • Remove salt and pepper noise • Minimize Gaussian noise • Medial filter and Weiner filter •
8.	**Removing noise using spatial filters** Write and execute programs to remove noise using spatial filters • Understand 1-D and 2-D convolution process • Use 3 X 3 mask for low pass filter and high pass filter •
9.	**Edge Detection** Write a program for edge detection using different edge detection mask.
10.	**Image Morphological Operations** Write and execute program for image morphological operations erosion and dilation